

Text Mining

Spring Semester 2025/2026

Project Assignment Handout {version 1.0}

This handout details the rules for the mandatory practical project for Text Mining, to be developed and completed during the academic calendar of the Text Mining class.

1. Project Summary

“Over time, major indexes go up and down based on internal and external factors. Performance like that excites investors, but typically in opposite ways. Constant gains lead some investors to expect more of the same. Others worry the good times are surely about to end. The former sentiment is sometimes called “bullish,” while the latter is referred to as “bearish.”¹

The goal of this project is to develop an NLP model capable of predicting Market sentiment based on tweets. In summary, with the NLP techniques you have learned during class, you must implement a classification model that receives tweets as inputs and is able to predict, for each tweet, if it describes a Bearish (0), Bullish (1), or Neutral (2) attitude.

The project should be developed using python 3 and libraries such as *NLTK*, *Scikit Learn*, *Hugging face* and *LangChain*. Also, the project can be solved in various ways, which means there is no exact correct solution.

2. Group Rules

The project should be done in a group of between one (1) to four (4) students; we consider this the ideal size for group work. Groups larger than four will NOT have project graded.

Important Project (partial work) Delivery Dates

Delivery Deadline: **midnight 15th of June 2026**

Project Report + Notebooks + Predictions

¹ <https://smartasset.com/financial-advisor/bullish-vs-bearish>



3. Project Starting Point – Corpora

The data for this project is divided in a file for training “train.csv”, and another file for testing “test.csv”:

- **Train** (9543 lines): Presents the tweets (“text”) and the sentiment label (“label”). Each tweet can have one of the following labels: Bearish (0), Bullish (1), or Neutral (2). You can divide this set in Train/Validation.
- **Test** (299 lines): The structure of these dataset is the same as the train set, except that it does not contain the “label” column. You are expected to provide the predicted status (0, 1 or 2) for each tweet in this set and **we will compare your predictions with the actual (true) labels**.

4. Solution Requirements

Your solution should present the following points:

1. **Data Exploration 2.00 pts**: Here you should analyze the corpora and provide some conclusions and visual information (bar charts, word clouds, etc.) that contextualize the data.
2. **Corpus split 0.50 pts**: You must apply some method to split your training corpus into train/validation sets to evaluate the performance of your model. You can also resort to K-Fold cross validation.
3. **Data Preprocessing 3.00 pts**: You must correctly implement at least four (4) of the data preprocessing techniques shown in class (stop words, regular expressions, lemmatization, stemming, etc.).
4. **Feature Engineering 5.50 pts**: You must correctly implement and experiment at least one variation of each of the following feature engineering techniques seen in class: BoW, word2vec, Transformer Encoder.
5. **Classification Models 4.50 pts**: You must correctly implement and test at least two variations of each the following classification methods seen in class, including Traditional ML (KNN, MLP, Logistic Regression, Random Forest, XGBoost, etc.) and Transformer Encoders.
6. **Evaluation and Analysis 1.50 pts**: You must correctly evaluate and compare your models resorting, at least, to Recall, Precision, Accuracy and F1-Score, and explain what the evaluation means in the context of the problem.

Moreover, the development of extra work (techniques not shown in class that are more advanced than the ones in point 5 above) is highly recommended and will account for a maximum of **2.00 points** as follows:

1. **Feature Engineering – 0.50 points** for each extra Transformer Encoder method applied (maximum of 2 extra methods).
2. **Classification Models – 1.00 points** for correctly using a decoder model for classification.



5. Delivery Guide

In terms of the solutions developed (see **delivery template folder that will be shared in Moodle**), you must deliver:

1. One .pynb file (notebook), named tm_tests_xx (xx stands for the group number), following the structure in section 4 of this handout and **containing the techniques you experimented with and their evaluation**.
2. Another .pynb file (notebook) named tm_final_xx with only your **ready-to-run final solution**. This solution should include a single pipeline with a single classification model.
3. A .csv file, named "pred_xx", with only two columns - **the id of the test set and your predicted labels for the test set**.

Additionally, you **must submit a PDF report** named "report_XX", documenting your work, with the following structure (other structures are also accepted):

1. **Data Exploration** – data presentation and explanation of the main finding from the exploratory analysis (accounts for **50%** of criteria **4.1**).
2. **Data Preprocessing** – explanation of the different preprocessing methods developed (accounts for **25%** of criteria **4.2** and **4.3**).
3. **Feature Engineering** – description and explanation of the feature engineering methods applied (accounts for **30%** of criteria **4.4**)
4. **Classification Models** – description and explanation of the models implemented (accounts for **30%** of criteria **4.5**)
5. **Evaluation and Results** – description of the performance of the models and main conclusions (accounts for **50%** of criteria **4.6**)

Extra Information:

- The PDF report should have a **maximum of 15 pages** describing the previous points. Exceeding this number will incur a **0.5-point penalty** for each extra page.
- Any **extra work** developed **must be clearly defined as such in the PDF report**, or else it will not be considered for evaluation as extra work!
- All files should be saved in a folder named "group_xx". This folder (zip it if you need) must be submitted through Moodle's project submission section until **23h:59 of the 23rd of January**.
- Failure to deliver on time will incur a **1.0-point penalty** for each half-day late.
- Failure to comply with the delivery guide above will meet with up to **1.0-point penalty**.
- To prevent plagiarism and misuse of Gen AI, students may be **randomly chosen for an oral defense** to assess their understanding.



6. Extra Challenge

Extra Challenge 1

We will compare your predictions with the actual Label from the test set ("test.csv").

The three (3) groups with the highest performance will receive points as follows:

- **1.00 points** for the group with the best model
- **0.50 points** for the group with the 2nd best model
- **0.25 points** for the group with the 3rd best model

Extra Challenge 2

We will consider up to **1.50 points** for groups that: for **correctly designing and implementing an agentic ai-based workflow that orchestrates the classification pipeline using tools or multiple models**. The agent must have a **conversational interface** that receives some prompt and performs at least one non-trivial decision or coordination task, such as:

- choosing between alternative models
- comparing outputs from multiple classifiers
- routing tweets to different models
- or automating evaluation

Simple use of a single LLM prompt for direct classification, without tool use, orchestration, or decision-making, will not be considered for this extra point. Likewise, to be considered for extra points, these developments must be described in detail in the report.

Final note: You may have noticed that, with the extra challenges, grades can hypothetically exceed 20 points; while that would be ideal, the final project grade is capped at 20, so any total above 20 will still be recorded as 20.

Questions and Clarifications

Should it be necessary, we will provide further clarifications and answers to questions from students, updating the respective Moodle page as appropriate.

Good luck with your project!